CAPPING WITH A CHERN CLASS IS WELL DEFINED ON RATIONAL EQUIVALENCE CLASSES OF CYCLES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as three spans, lines 8-12 (the verbatim package together with the project's own reference, slogan and history display-environment definitions, including the author's commented-out older form of the reference environment), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). The lines 8-12 span is required because a counted statement or its surrounding text may use those display environments, which the project typesets as comments. Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- Uniqueness of the classes obtained by capping with the Chern classes, determined by a single equation on the projective bundle. Same chapter chow.tex, label lemma-determine-intersections (invoked at proof line 7374). It shows that the collections $\alpha_j = c_j(\mathcal{E}) \cap \alpha$ and $\beta_j = c_j(\mathcal{E}) \cap \beta$ are each uniquely determined by the same equation in the chow group of $\mathbf{P}(\mathcal{E})$.
- Compatibility of flat pullback with rational equivalence. Same chapter chow.tex, label lemma-flat-pullback-rational-equivalence (invoked at proof line 7380). It guarantees that the images of the rationally equivalent cycles α and β in the projective bundle satisfy the same equation, so that the uniqueness applies.

Source reference. Original lemma 02U6: sources/stacks-chow/pinned-original-source.tex, source lines 7280–7280. Navigation pointer for \ref{lemma-determine-intersections} used inside the counted span (line 7374); the argument that consumes it is retained in the span.

Source reference. Original lemma 02S1: sources/stacks-chow/pinned-original-source.tex, source lines 3173–3173. Navigation pointer for \ref{lemma-flat-pullback-rational-equivalence} used inside the counted span (line 7380); the argument that consumes it is retained in the span.

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Source reference. Original reference 02QL: sources/stacks-chow/pinned-original-source.tex, source lines 1372–1372. Navigation pointer for \ref{situation-setup} used inside the counted span (line 7365); the argument that consumes it is retained in the span.

Lemma 0.1. *Let (S, δ) be as in Situation 02QL. Let X be locally of finite type over S. Let $\mathcal{E}$ be a finite locally free sheaf of rank r on X. If $\alpha \sim_{rat} \beta$ are rationally equivalent k-cycles on X then $c_j(\mathcal{E}) \cap \alpha = c_j(\mathcal{E}) \cap \beta$ in $\mathrm{CH}_{k-j}(X)$.*

Proof. By Lemma 02U6 the elements $\alpha_j = c_j(\mathcal{E}) \cap \alpha$, $j \geq 1$ and $\beta_j = c_j(\mathcal{E}) \cap \beta$, $j \geq 1$ are uniquely determined by the *same* equation in the chow group of the projective bundle associated to $\mathcal{E}$. (This of course relies on the fact that flat pullback is compatible with rational equivalence, see Lemma 02S1.) Hence they are equal. □